

Moore-Penrose pseudoinverse applied to medical digital images restoration

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1 Introduction

The scope of this project is to study an application of the Moore-Penrose pseudoinverse. In particular, taking inspiration from the research article [1], an application on medical X-ray images has been investigated: some images may suffer from blurring due to movement, and in a similar context, it is very important to have a steady image in order to detect fractures.

The choice of this application field was triggered by two main facts: I have a bachelor's degree in Physiotherapy and worked for some years in that field; almost 1 year ago, I had a small fracture of the cuboid bone (just like the picture in this case, although not mine).

The code has been developed using Matlab.

2 Dataset

An image referring to a cuboid fracture (bone of the foot) has been used as a dataset, with dimensions of 482×284 pixels (height $\times$ width) and 3 channels.

The image has been uploaded using the `imread` function provided by MATLAB and then transformed into grayscale, reducing the number of channels to work on a matrix.

Furthermore, the values associated with each pixel were divided by the maximum value present to scale all values into the interval $[0, 1]$.

The original image is displayed in Figure 1.

3 Image degradation

In order to simulate the effect of blurring caused by a linear motion, it is assumed that such motion corresponds to a discrete number of pixels and is aligned with the horizontal or vertical sampling.

For the scope of this project, it was assumed that the matrix corresponding to the blurring effect was retrieved using certain methods, and it looks like the one displayed here below:

$$H = \begin{bmatrix} k_1 & \cdots & k_n & 0 & 0 & 0 & 0 \\ 0 & k_1 & \cdots & k_n & 0 & 0 & 0 \\ 0 & 0 & k_1 & \cdots & k_n & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & k_1 & \cdots & k_n \end{bmatrix}$$

The matrix $\mathbf{H}$ has a shape of $r \times m$ and is defined as follows: the elements $k_1, k_2, \dots, k_n$ are defined as $k_l = \frac{1}{n}$ for $1 \leq l \leq n$, where n indicates the linear motion blur in pixels.

The degradation matrix may also be defined as:

$$x_{\text{out}_i} = \frac{1}{n} \sum_{h=0}^{n-1} x_{\text{in}}(i+h)$$

representing an undetermined system with r simultaneous equations and $m = r + n - 1$ unknown variables.

The degraded image is computed by performing column by column the following multiplication:

$$x_{\text{out}} = \mathbf{H} \cdot x_{\text{in}}$$

where x_{in} is an individual column of the original image, and x_{out} is the corresponding one in the degraded image.

The columns are then horizontally stacked, resulting in a degraded image of size 478×284 , with a reduction in the number of vertical pixels.

This image is displayed in Figure 2.

4 Restoration of original image

The idea of restoring the image starts from the following equation:

$$x_{\text{in}} = H^{-1} \cdot x_{\text{out}}$$

Therefore, it is necessary to compute a pseudoinverse of the rectangular matrix $\mathbf{H}$.

The Moore-Penrose pseudoinverse (also known as the generalized inverse) of a matrix A , denoted as A^+ , is a powerful tool for solving underdetermined systems. It allows us to find a solution that minimizes the Euclidean norm of the error vector $\|Ax - b\|_2$. Given a matrix $A \in R^{m \times n}$, the pseudoinverse $A^+ \in R^{n \times m}$ is a matrix satisfying the four Moore-Penrose conditions:

1. $A A^+ A = A$
2. $A^+ A A^+ = A^+$
3. $(A^* A^+)^T = A^* A^+$
4. $(A^+ A)^T = A^+ A$

With the pseudoinverse at hand, we can find a solution for $\mathbf{x}$ as follows:

$$\mathbf{x} = \mathbf{A}^+ \mathbf{b}$$

This solution yields the least squares solution, which minimizes the sum of squared errors. While it may not be unique, it represents the best approximation of $\mathbf{x}$ given the underdetermined system.

In the project, the pseudoinverse is computed using the singular value decomposition (SVD), as follows:

1. First, the singular value decomposition of $\mathbf{H}$ is computed using the `svd` function provided by MATLAB:

$$\mathbf{H} = \mathbf{U} \Sigma \mathbf{V}^T$$

The columns of $\mathbf{U}$ corresponding to non-null singular values (the entries on the diagonal of Σ) form a basis of the range of $\mathbf{H}$, while the columns of $\mathbf{V}$ (i.e., rows of $\mathbf{V}^T$) corresponding to null singular values form a basis for the kernel of $\mathbf{H}$.

2. Then, the Σ^+ matrix $\in R^{n \times n}$ is assembled as follows:

$$(\Sigma^+)_{ij} = \begin{cases} \frac{1}{\sigma_i} & \text{for } i = j = 1, \dots, r \\ 0 & \text{for } i = j = r + 1, \dots, n \text{ or when } i \neq j \end{cases}$$

where r is the rank of $\mathbf{H}$. Basically, it takes only the reciprocal of the non-(numerically) null singular values.

3. Finally, $\mathbf{A}^+$ is computed by the following matrix multiplication:

$$\mathbf{A}^+ = \mathbf{V} \Sigma^+ \mathbf{U}^T$$

It can be deduced that the computational cost is mainly related to the SVD computation step.

5 Techincal details of the project

In the experiment, the degradation was computed by setting $n = 10$, leading to $H \in R^{473 \times 482}$, with a rank of 473 (full rank).

This resulted in a Σ^+ matrix without any null values on the diagonal after the application of SVD (also providing as tolerance $1e - 12$).

The pseudoinverse computed with the implemented method was compared with the one provided by the MATLAB method `pinv`, resulting in a norm of the difference equal to 2.6986×10^{-13} , indicating that the custom method is a good approximation.

The computed H^+ was tested for the four Moore-Penrose properties, providing positive results.

Finally, the reconstructed image was computed by applying the pseudoinverse matrix to every column of the degraded image.

Figure 3 X shows the results of the reconstruction procedure. The image is close to the original image, with details well visible (note the horizontal bands related to the horizontal uniform blurring movement).

For a quantitative analysis, the norm of the distance between the original and restored image was computed, resulting in a value of 405.1782. While this is a relatively high value, it's important to note that in this field of application, the quality of the image has been significantly improved, allowing an expert eye to evaluate the radiography.

This concludes this project, which aimed to investigate the application of the Moore-Penrose pseudoinverse in a real-world use case.

(1) Spiros Chountasis, Vasilios N. Katsikis, and Dimitrios Pappas, "Applications of the Moore-Penrose Inverse in Digital Image Restoration", available at: <https://eudml.org/doc/45799>

Original



Figure 1: Original image of an foot Xray.



Figure 2: Degraded image of an foot Xray with $n=10$ (blur effect).



Figure 3: Degraded image of an foot Xray with $n=10$ (blur effect).